Harshwardhan Fartale

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Work Experience

ZS

Advanced Data Science Associate — *October 2025 - Present*

- Built an agentic AI platform for life sciences, automating FDA dossier reviews with word-level traceability across 2,000+ source documents.
- Built and scaled a custom semantic LLM prompt router to handle 3,000+ requests daily.
- Post-trained small language models for healthcare, achieving performance comparable to frontier models.

DRDO & (IISC) Indian Institute of Science, Bangalore

Predoc Associate — *July 2024 - September 2025*

- Wrote C++ inference modules for CASDIC combat-aircraft hardware, optimizing ML model execution under embedded latency constraints and reducing inference time by 35% versus baseline.
- Fine-tuned IBM Granite and Gemma models for scientific ML tasks, built a Graph-RAG pipeline that outperformed baselines by 45%, and led development of **PULSE** and **SciREX** under **AiREX Lab**.

Bajaj Finserv Health

Data Scientist — *Jan 2024 - June 2024*

- Built a fraud detection model that caught 20% more abusive transactions than the existing rule-based system, processing 3000+ transactions/day across Bajaj Health's payments pipeline.
- Built a graph-based fraud detection pipeline in Neo4j and Memgraph with optimized SQL feature extraction, improving linked-fraud recall by 55% and model precision by 30% versus tabular baselines.

OpenMined

Open Source Technical Writer — *Jan 2023 - June 2023*

- Wrote beginner tutorials and technical documentation on differential privacy and federated learning for PySyft, and integrated them into OpenMined's onboarding docs.
- Published blog posts explaining privacy-preserving ML concepts, driving monthly reads across OpenMined channels.

Publications

GPU Programming with Triton
Manning Publications — Author (MEAP, 2026)

Can Linguistically Related Languages Guide LLM Translation in Low-Resource Settings?
LoResMT: EACL Workshop (2026)

The Evolution of FlashAttention
ICLR Blog Post Track (peer-reviewed) (2026)

Disentangling Recall and Reasoning in Transformer Models through Layer-wise Attention and Activation Analysis
AAAI XAI4Science Workshop (2026)

RADAR: Mechanistic Pathways for Detecting Data Contamination in LLM Evaluation
NeurIPS LLM Evaluation Workshop (2025)

Equilibrium Dynamics and Mitigation of Gender Bias in Synthetically Generated Data Evaluation
AAAI Workshop on Shaping Responsible Synthetic Data in the Era of Foundation Models (2025)

Skills

Languages	Python, C++, Go, SQL, Triton
ML/AI	PyTorch, Scikit-learn, LLM pre-training & post-training, RL, RSI, Agentic AI, fine-tuning, Graph-RAG
Infrastructure	Docker, Kubernetes, GCP, FastAPI, inference optimization, GPU, Neo4j, Memgraph, PostgreSQL
Technical Writing	**GPU Programming with Triton** ; **LessWrong** , **Medium** , **Hashnode** , **Substack** , **Writings**

Key Projects

Chalk - AI Agent RL Environments

- Built Chalk, an open-source framework for AI agent RL environments across coding, math, and custom tasks, with isolated execution and automated reward scoring.

Education

National Institute of Technology, Hamirpur — *CGPI – 8.56*

B.Tech in Electrical Engineering, GATE Data Science (DA) AIR 663 — *Dec 2020 – May 2024*

Additional Experience

IISc Bangalore

Teaching Assistant, DS-246 Generative & Agentic AI in Practice — *Jul 2025 – Sep 2025*

DRDO Industry Academia Centres of Excellence (DIA-CoEs)

ML & Data Science Trainer for DRDO Scientists — *Jan 2025*